Pearson

GCSE Computer Science: Effective Teaching and Learning for Paper 2

Welcome to this Professional Development Training

This free online event supports teachers delivering Pearson Edexcel GCSE Computer Science whether you are new or have been teaching this course for five years. It is part of a series of connected webinars this school year.

The training covers a variety of approaches for the effective teaching of Paper 2 and how to provide feedback to students on their progress.

You will:

- be shown options for balancing the theory and programming elements of the course
- see how to organise the existing Pearson CT (Computational Thinking) units with free online platforms and resources
- be shown practical approaches for T&L of programming
- consider forms of assessment and feedback in both Years 10 and 11.

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Questions

Please answer the following poll questions when they are launched:

- When was / is your first Pearson GCSE Computer Science cohort exam?
- What platform does your school use?

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Agenda

Effective Teaching and Learning for Paper 2, 60mins

In this session we are going to:

- **share the different contexts** that we have in our schools
- share the **key documents** you should know fully
- consider the best **balance for teaching theory and programming** although this may vary between classes
- look at resources to teach programming starting with the **free Pearson Edexcel units** but also ways to enhance with free **additional resources**
- look at **practical approaches for Teaching and Learning programming** including the recent **feedback from the examiners**
- show examples of **assessment in Year 10 and Year 11** – it might vary…
- **future webinar dates and Subject Advisor contact.**

Questions

Please answer the following poll questions when they are launched:

- Which IDE(s) do you use in the exam?
- Which IDE(s) do you use for regular teaching?

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Key documents for Paper 2

GCSE (9–1) Computer Science Specification (p5, 13–17, –) – link

Programming Language Subset (PLS) Booklet (*v6 / coloured paper)* – link*

Pearson Edexcel's Good Programming Practice Guide (Issue 1.2) – link*

Instructions for Conduct of the Examination (ICE 1.9) – link*

Pearson Edexcel's Getting Started Guide (Issue 3, p5–8, 9, 26 – Topic 6) – link*

Specimen Exam papers / SAMs 1–4 / 2022–2025 Exam past papers – link

**Always check you have the latest versions.*

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What is your balance of teaching theory and practical?

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Questions

Please answer the following poll questions when they are launched:

- What is your **curriculum time allocation *per fortnight*** in hours?
- How do you **balance your lesson time for both theory and programming** ?

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A1. Programming must be done…

Page 16 of the specification reminds us that we all have to do some programming…

… and the Practical Programming Statement needs to be signed by SLT and sent by 31st May each year.

Practical Programming Statement (PPS)

Centres are required to complete a Practical Programming Statement (PPS), which confirms they have taken reasonable steps to ensure that each student sitting our GCSE in Computer Science has had the opportunity to demonstrate their programming skills in terms of design, write, test and refine programs in Python during their course of study. The PPS should be completed by a member of the senior leadership team at the centre. Failure by a centre to provide a completed PPS to us in a timely manner will result in potential malpractice and/or maladministration.

The PPS will be made available by the 31st of January in the year of the examination on the GCSE in Computer Science section of our website.

The completed PPS must be submitted to us by the 31st of May in the year of the examination. Centres are reminded that if they do not submit this form it may result in malpractice/maladministration.

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A2a. Balance – WAGOLL 1

Block 6	Y10_06_P	Y10-06-PSum1	P31	LANs and WANs	Give reasons why computers are connected on a network Differentiate between a LAN and a WAN Categorise tasks according to the type of network used to carry them out Explain the benefits to organisations of a WAN Explain why protocols are needed on a network Describe the purpose of an IP address	4.1.1 4.1.2 4.1.3 4.1.6	29.04.2025
			P32	Network speed	Define the meanings of the terms 'bandwidth' and 'latency' Explain how bandwidth and latency affect the performance of a network Use bits per second (bps) to describe network speed Construct expressions involving file size, transmission rate and time	2.3.1 4.1.5	01.05.2025
			P33	Connectivity	Differentiate between wired and wireless connectivity Explain how data are transmitted along copper and fibre-optic cables Compare the performance of copper and fibre-optic cables and give examples of their use Describe how high-speed broadband is delivered	4.1.4	13.05.2025
			P34	Wired vs wireless	Describe how devices are connected on a wireless network Compare the performance of wired and wireless LANs and give examples of situations where one is preferable to the other Summarise the characteristic of Wi-Fi, Bluetooth, RFiD, Zigbee and NFC and give examples of their use	4.1.4	15.05.2025
			P35	Network topologies	Define the term 'topology' Describes the characteristics of bus, star and mesh network topologies Draw and label a diagram of each topology Match descriptions to network topologies Match descriptions of what they do to internet components (backbone, POP, NAP, router)	4.1.3 4.1.8	03.06.2025
			P36	Assessment	Give three reasons for connecting devices in a network Explain how a LAN differs from a WAN Define the term 'internet backbone' Describe the function of a router Explain how data are transmitted on a fibre-optic cable State two advantages and two disadvantages of using wireless to connect devices on a LAN rather than cable Construct an expression to calculate the time needed to transmit a file over a network Explain why protocols are needed on a network	4.1.1 4.1.2 4.1.3 4.1.4 4.1.5 4.1.6	05.06.2025

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A2b. Balance – WAGOLL 1

Block	Section	Subfolder	Lesson number	Lesson title	What students will learn	Spec. ref.	Date
Block 6	Y10_06_CT	Y10-06-CTSum1	CT31	Turtle introduction, pens and lines	Decompose a problem Use turtle graphics to draw lines	6.1.1 6.1.2 6.1.4	24.04.25
			CT32	Turtle movement, coordinates, polygons, subprograms	Use Cartesian coordinates Incorporate selection, repetition, and iteration into turtle graphics Use subprograms	6.2.2 6.3.1 6.3.2 6.4.1 6.6.2	24.04.25
			CT33	Turtle pens, colours, filling, and circle	Use turtle pens of different colours Use turtle pens of different sizes Use turtle fill in closed shapes	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2	08.05.2025
			CT34	Turtle, combining subprograms, layers	Combine subprograms to produce a turtle graphics image	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2	08.05.2025
			CT35	Turtle big problem	Decompose a problem into smaller parts Combine subprograms to create a solution	1.1.1 6.1.1 6.1.2 6.4.1 6.6.1 6.6.2	22.05.2025
			CT36	Assessment	Use the turtle module, programming constructs, and subprograms to create images	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2	12.06.2025

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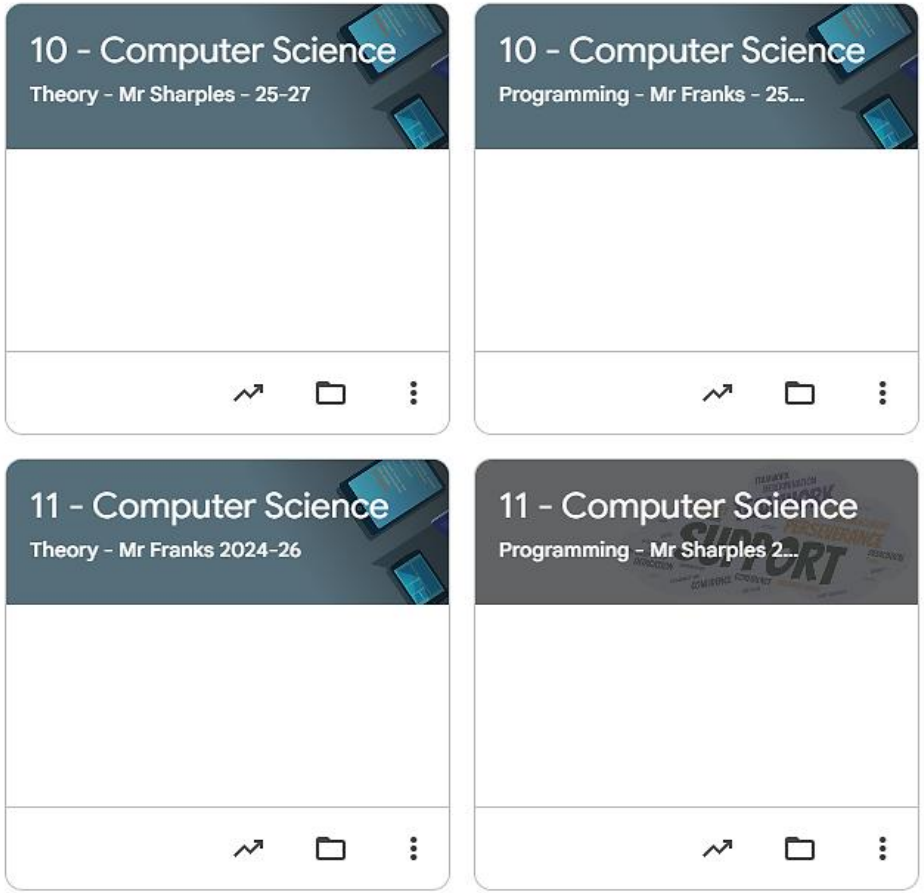
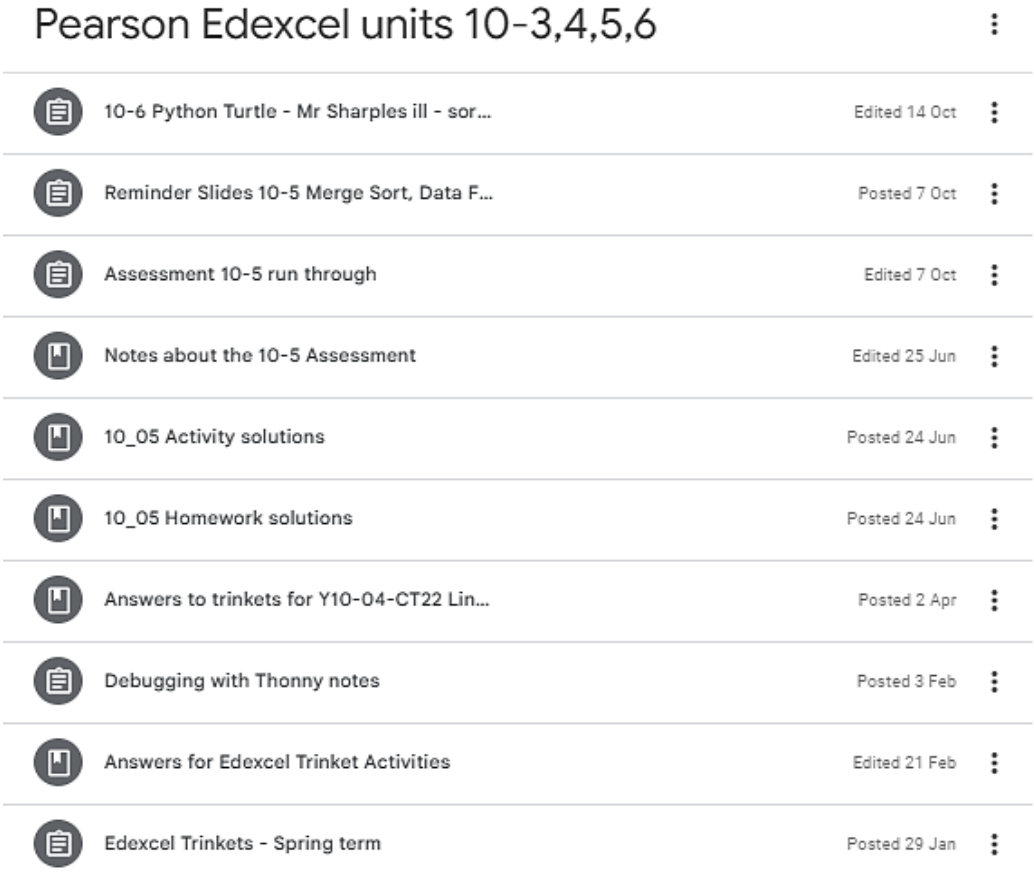
A3a. Balance – WAGOLL 2 Using Google Classrooms

Separate Google Classrooms for theory and programming through Years 10 and 11.





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A3b. Balance – WAGOLL 2 in detail

1 lesson (55m) Theory per week, 2 lessons (a double – 1h 50m) Programming per week:

	2024-26	Mr Franks	Mr Sharples	Mr Sharples
	Blank Class	Mon p7 Theory	Tues p1 Programming	Tues p2 Programming
01/09/2025				
08/09/2025		*5.1 - Environmental issues - PGO slides*	GPPG Booklet 1	Revision
15/09/2025		Smart Revise - reminder	Topic test 10-5 Files and Authentication + Qu6	Topic test 10-5 Files and Authentication + Qu6
22/09/2025		GPPG Booklet 2 / SR time	Go through reading files	Go through reading files
29/09/2025		5.3 Legislation and privacy RPI lesson & PGO slides	Go through writing files	Go through writing files
06/10/2025		GPPG Booklet 3 / SR time	Go through authentication	Go through authentication
13/10/2025		GPPG Booklet 4 / SR time	T2Code 10 - Turtle	T2Code 10 - Turtle
20/10/2025	*Halfterm*	Revision / SR time	Revision / SR time	Revision / SR time
27/10/2025	*Halfterm*	Revision / SR time	Revision / SR time	Revision / SR time
03/11/2025		*Inset day*	***Mocks***	***Mocks***
10/11/2025		***Mocks***	***Mocks CS Paper 1 - SAM04***	***Mocks CS Paper 1 - SAM04***

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A4. Analysis –

How do you know if the balance of teaching Theory: Programming works for your class or classes?

Pearson have looked internally every year to see if there is any significant difference across the whole cohort between P1 and P2 and they do not see any significant difference in outcomes.

However, you can do your own analysis each year to see if there is a difference:

I think there is a clear pattern for the majority of students that 1:2 Theory: Programming time in lessons results in better programmers…

Sample selection from real 2025 results:

P	Q	R	S	T	U	V	W
	Paper 1 /75		Paper 2 /75		residual to show difference in performance in theory cf programming	Total both /150	Total both %
pupil 1	58	**77%**	75	**100%**	-23%	**133**	**88.67**
pupil 2	7	9%	8	**11%**	-1%	**15**	**10.00**
pupil 3	46	**61%**	38	**51%**	11%	**84**	**56.00**
pupil 4	27	36%	31	**41%**	-5%	**58**	**38.67**
pupil 5	31	41%	64	**85%**	-44%	**95**	**63.33**
pupil 6	27	**36%**	34	**45%**	-9%	**61**	**40.67**
pupil 7	52	**69%**	65	**87%**	-17%	**117**	**78.00**
pupil 8	28	**37%**	32	**43%**	-5%	**60**	**40.00**
Averages:		46%	43	58%	-12%	**43**	**28.92**

Column U shows the comparison in results /75 for P1 and P2. Negative value favours programming.

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A5. Other considerations

Any of these may be relevant to your balancing decision:

- Ability profile of students
- Motivation
- Time of day
- Availability of teachers
- Prior experience of teacher AND students
- Teacher led / independent working / paired working
- The progress of each class

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Resources to teach programming

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B1a. Resources to teach programming

Scheme of work - Learning content mapping

The materials of our Scheme of Work have been classified into blocks, from 1 to 10.
You can download the materials for each block individually as a zip folder.

	Term1		Term 2		Term 3	
	Half Term 1	Half Term 2	Half Term 1	Half Term 2	Half Term 1	Half Term 2
Year 10	Block 1	Block 2	Block 3	Block 4	Block 5	Block 6
Year 11	Block 7	Block 8	Block 9	Block 10		

Scheme of work

GCSE (9-1) Computer Science Scheme of Work Guidance
| PDF 1.3 MB | 07 June 2024

GCSE (9-1) Computer Science Introduction to the Scheme of Work
| PDF 1.0 MB | 05 June 2024

GCSE (9-1) Computer Science Scheme of Work - Learning Content Mapping
| XLSX 367.0 KB | 05 June 2024

GCSE (9-1) Computer Science Scheme of Work Block 1
| ZIP 11.6 MB | 05 June 2024

Download content for each of the 10 blocks

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B1b. Resources to teach programming

Lesson number	Lesson title	What students will learn	Spec. ref.
CT31	Turtle introduction, pens and lines	Decompose a problem Use turtle graphics to draw lines	6.1.1 6.1.2 6.1.4
CT32	Turtle movement, coordinates, polygons, subprograms	Use Cartesian coordinates Incorporate selection, repetition, and iteration into turtle graphics Use subprograms	6.2.2 6.3.1 6.3.2 6.4.1 6.6.2
CT33	Turtle pens, colours, filling, and circle	Use turtle pens of different colours Use turtle pens of different sizes Use turtle fill in closed shapes	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2
CT34	Turtle, combining subprograms, layers	Combine subprograms to produce a turtle graphics image	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2
CT35	Turtle big problem	Decompose a problem into smaller parts Combine subprograms to create a solution	1.1.1 6.1.1 6.1.2 6.4.1 6.6.1 6.6.2
CT36	Assessment	Use the turtle module, programming constructs, and subprograms to create images	6.2.2 6.3.1 6.3.2 6.6.1 6.6.2

- CT33_Code_Snippets
- CT33_Slide_Code_Snippets
- Y10_06_CT33_Activities
- Y10_06_CT33_Activities_Solutions
- Y10_06_CT33_Homework
- Y10_06_CT33_Homework_Solutions
- Y10_06_CT33_LessonPlan
- Y10_06_CT33_Slides

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B1c. Resources to teach programming

Activity 2

Here is a Python program.

```
# Program to calculate the volume of a cone
PI = 3.141  # create a constant for the value of PI
radius = 5
print("Volume of the cone is", volume)
volume = PI * radiusSquared * (height/3)
radiusSquared = radius ** 2
height = 8
```

- Copy and paste the program into your Integrated Development Environment (IDE) and run it.

- Re-order the lines of code so that the program runs without error.

.....

.....

.....

.....

.....

.....

Describe what happens when you run the program provided.	

[Pearson's Good Programming Practice Guide]

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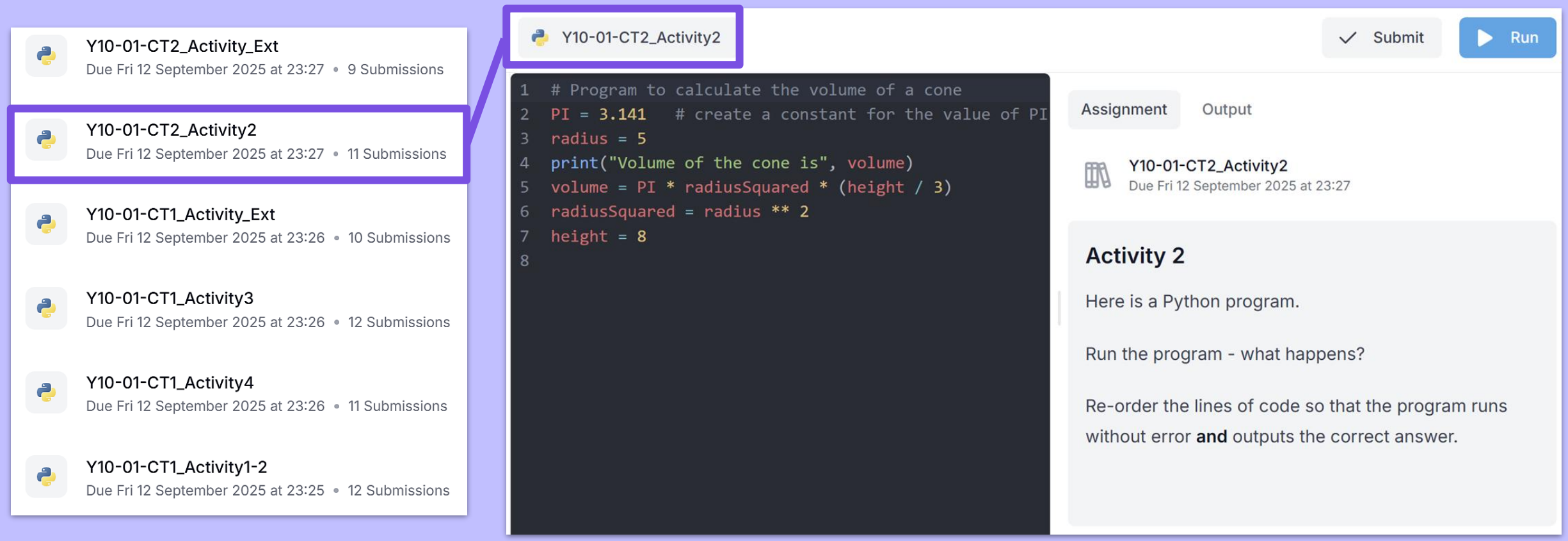
B1d. Resources to teach programming



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B2a. Resources to teach programming – Trinket Connect

trinket Connect | Home | Plans | Learn | Help | csharplesadmin | Sign Out

@RS Pearson Edexcel Y10/11 Trinkets THIS ONE 2025-26 by Chris Sharples

Dashboard | Course | Users

- Y10-04-CT24: Assessment Qu6*
- Y10-04-CT24: Assessment Qu8*
- Y10-04-CT24: Assessment Qu9 (PGO2A Qu...

Page Assignment

10-05

- Y10-05-CT25: Merge sort MISS OUT 2025
- Y10-05-CT26: Reading files*
- Y10-05-CT27: String processing*
- Y10-05-CT28: Writing files*
- Y10-05-CT29: Authentication*
- Y10-05-CT26272829Homework answers
- Y10-05-CT30: Assessment Qu3
- Y10-05-CT30: Assessment Qu4
- Y10-05-CT30: Assessment Qu5

Page Assignment

10-06

- Y10-06-CT31: Turtle introduction, pens an...
- Y10-06-CT32: Turtle movement, coordinat...
- Y10-06-CT33: Turtle pens, colours, filling a...
- Y10-06-CT34: Turtle combining subprogra...
- Y10-06-CT35: Turtle big problem
- Y10-06-CT36: Assessment02
- Y10-06-CT36: Assessment03

Topic | Outline

Y10-05-CT26: Reading files* | Edit

Click here for Slides for Y10-05-CT26: Reading files

Worksheet Activities followed by Homework Tasks.

Y10_05_CT26_Activities:

Activity 1a - Reading and just printing data from a file

A program has been started that loads data from a file containing records. Use slide 10 to read the data into your python code and print it out.

main.py | CT26_Activity1.txt | Save

```
#Y10_05_26_A1Remix2025

# ------------------------------------------------------------
# Constants
# ------------------------------------------------------------
FILENAME = "CT26_Activity1.txt"

# ------------------------------------------------------------
# Global variables
# ------------------------------------------------------------
usersTable = []                    # Holds all the records

# ------------------------------------------------------------
# Subprograms
# ------------------------------------------------------------
def loadData ():

    # =====> Open the file for reading only
    theFile =

    # Use the example code in Slide 10 to output  out the data

    # =====> Close the file
```

Result | Instructions

A program has been started that loads data from a file containing records.

Use slide 10 to read the data into your python code and print it out.

Activity 1b - Reading data from a file to create a list

main.py | CT26_Activity1.txt | Save

```
#Y10_05_26_A1Remix2025b-----------------------------------------
# Constants
# ------------------------------------------------------------
FILENAME = "CT26_Activity1.txt"

# ------------------------------------------------------------
# Global variables
```

Result | Instructions

A program has been started that loads data from a file containing records.

Amend the code to complete the program.

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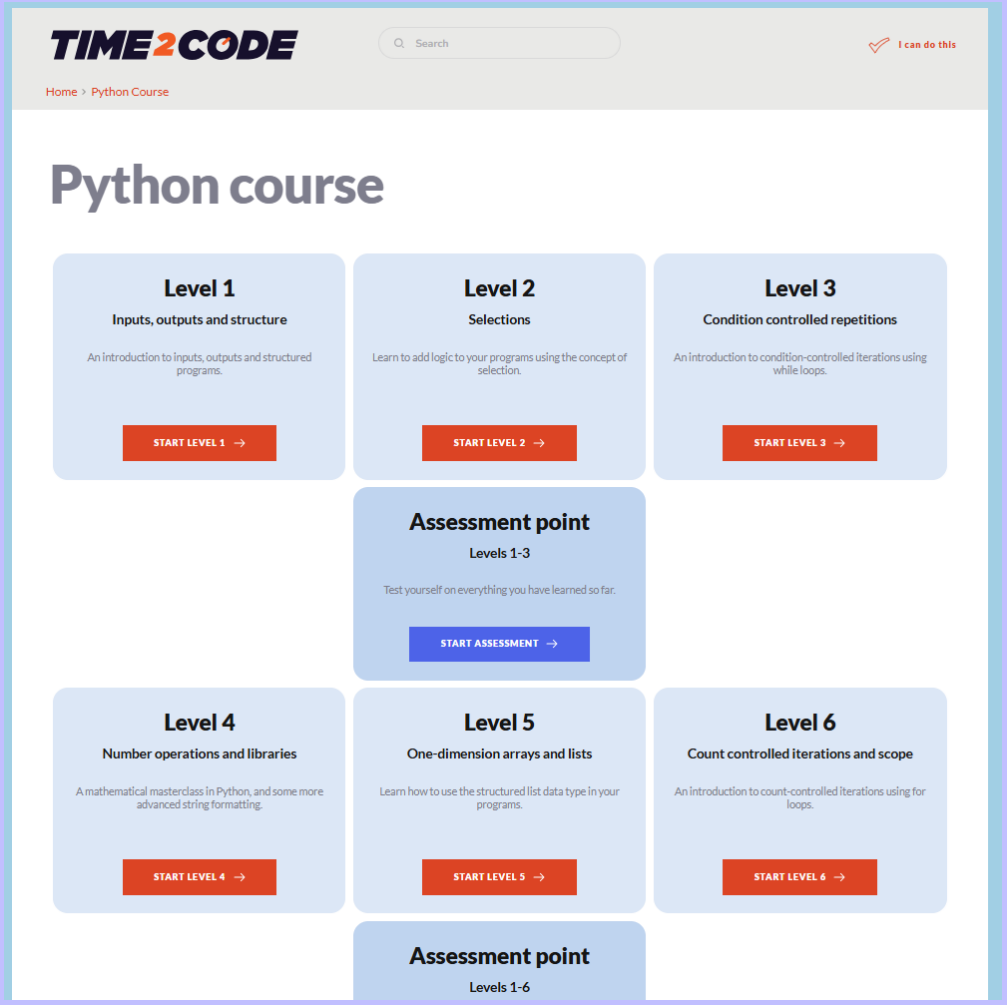
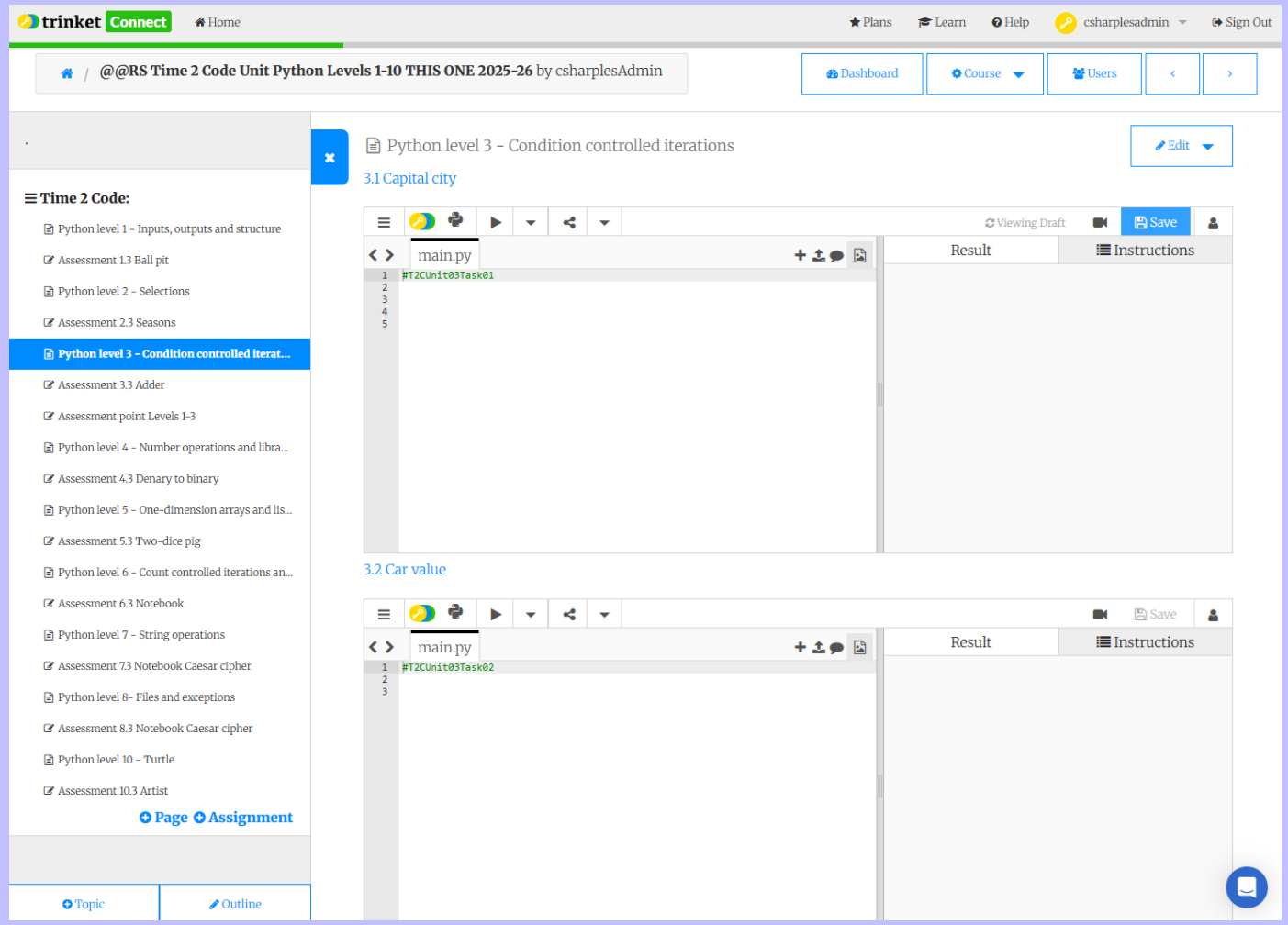
B2b. Time 2 Code – Craig’n’Dave – Free

CRAIG N DAVE





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B2c. Google Colab (integrated Jupyter notebooks)

10-5 Assessment run through.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Connect Share

Work through each question to get a working programme.

Question 3

A program is needed to write the contents of a two-dimensional data structure to an output file. Open 'CT30_Q3_Student.py' into your development environment.

Amend the code to write the contents of the two-dimensional data structure, 'towns', to the comma-separated value text file named 'towns.txt'.

The output file should have exactly 10 lines.

```
# ------------------------------------------------------------
# Import libraries
# ------------------------------------------------------------

# ------------------------------------------------------------
# Constants
# ------------------------------------------------------------
FILENAME = "towns.txt"
LF = "\n"


# ------------------------------------------------------------
# Global variables
# ------------------------------------------------------------

towns = [["Bournemouth","Dorset",198727],
         ["Bath","Somerset",101106],
         ["Reading","Berkshire",229815],
         ["Plymouth","Devon",241179],
         ["Portsmouth","Hampshire",248479],
         ["Brighton","East Sussex",241999],
         ["Worthing","West Sussex",114212],
         ["Southend-onSea","Essex",183809],
```

Pearson Edexcel units 10-3,4,5,6

10-6 Python Turtle - Mr Sharples ill - sor... Edited 14 Oct

Reminder Slides 10-5 Merge Sort, Data F... Posted 7 Oct

Assessment 10-5 run through Edited 7 Oct

No due date

link for later
https://docs.google.com/document/d/1KrkfD8LXMvfrsAz6pJgOiJ56uyDLLjB4Qg34Dj
usp=sharing

0 Handed in

8 Assigned

CT30_Q6_data.txt
Text

lakes.txt
Text

Sign in - Google Accoun...
https://colab.research.google.c

Y10_O5_CT30_Asst_Sol...
Google Docs

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Summary of free resources to teach programming

Here is a list of the free resources mentioned:

- Pearson/Edexcel Computer Science – Scheme of Work resources – link
- Edublocks Classrooms – link
- Craig'n'Dave's Time 2 Code – link
- Google Colab (hosted Open Source Jupyter notebooks) – link
- Pete Dring's Blog with Code – link
- Trinket.io (Trinkets are free to set up but the editable course version (Connect) needs a licence) – link

Practical approaches for T&L of programming

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C1. June 2025 Chief Examiner Findings

Paper 2 issues:

- Students struggle to run, test and debug programs effectively.
- Students fail to use [*read/write*] scaffolding and comments effectively.
- Many learners struggle with layout, indentation and sequencing logic.
- Students often gain partial functionality but fail to refine or optimise code.
- Students struggle to plan, break down and structure programs.

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Practical Strategies for Teaching Programming (Part 1)

Building confident, independent coders:

- Use PRIMM or Use–Modify–Create frameworks
- Make testing a habit
- Embed deliberate debugging practice
- Employ scaffolded coding and fade support
- Promote collaboration and explanation

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Practical Strategies for Teaching Programming (Part 2)

Developing structured, efficient and decomposed solutions:

- Use Parsons problems regularly
- Focus on structure before syntax
- Model iterative improvement
- Teach decomposition through refactoring
- Compare and critique code

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C2. Understanding the structure of Paper 2

- Focus on reading, analysing and refining code
- Linked to Python 3 content in the PLS
- Questions test analysis and refinement, not just writing from scratch
- Typical tasks: fix errors, rearrange code, improve readability, select correct lines
- Final question = full program design (covered in June's webinar)
- Apply standard ways of working – clear, consistent structure

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Suggested time: 20 minutes

3 A program allows the user to enter a string.

If the string is fewer than three characters long, an error message is displayed and the string is not processed.

Otherwise, the program adds up the ASCII values for each character in the string.

When the total is divisible by four, a four is added to the string.

When the total is divisible by five, a five is added to the string.

When the total is divisible by both four and five, a four is added to the string.

When the total is not divisible by either four or five, a zero is added to the string.

The program displays the total value and the changed string.

The table shows accurate test results for six inputs.

Input	Output
X	String must be three or more characters long
XY	String must be three or more characters long
cup	328 cup4
snow	455 snow5
lan	280 lan4
HAT	221 HAT0
<empty>	No output, but exits program

The lines of code in the program are mixed up. The indentation has been done for you.

PRIMM & Structure Awareness
Students familiar with PRIMM or Use–Modify–Create will already be used to *reading, predicting and reasoning* about given code before editing it.

Parsons & Scaffolding Practice
Regular Parsons-style classroom tasks train students to recognise structure, indentation and sequencing — exactly the skill needed to re-order these mixed lines correctly.

Testing & Debugging Routines Predict–Run–Compare routines help learners trace logic and verify output step-by-step, matching given test data.

Q03.py - Notepad

File Edit Format View Help

```
# ---------------------------------------------------------------
# Constants
# ---------------------------------------------------------------


# ---------------------------------------------------------------
# Global variables
# ---------------------------------------------------------------


# ---------------------------------------------------------------
# Main program
# ---------------------------------------------------------------

        print ("String must be three or more characters long")
            total = total + asciiValue
        # Check if divisible by 4, 5, or neither
        if (total % FOUR == 0):
            inString = inString + str (ZERO)
    if (len (inString) >= THREE):
ZERO = 0
THREE = 3
        else:
inString = ""
        for character in inString:
total = 0
            inString = inString + str (FIVE)
        total = 0                        # Reset each pass
        elif (total % FIVE == 0):
asciiValue = 0

FOUR = 4
    else:
FIVE = 5
while (inString != ""):
        print (total, inString)
            asciiValue = ord(character)

    inString = input ("Enter a string (blank to exit): ")
# Set up the input loop
inString = input ("Enter a string (blank to exit): ")
            inString = inString + str (FOUR)
```

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Suggested time: 25 minutes

4 A program is required to calculate the values in a multiplication square.

Here is an example of a multiplication square.

×	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15
4	4	8	12	16	20
5	5	10	15	20	25

$2 \times 5 = 10$

The number 10, at the intersection of row 2 and column 5, is the result of 2×5

The user enters a number for the sides of the square into the program.

The program has these requirements:

- calculate the values for the multiplication square
- write the values to a new file
- separate values with commas
- do not include the row and column labels.

Here are the contents of the output file, when the user enters the number 5

```
1,2,3,4,5
2,4,6,8,10
3,6,9,12,15
4,8,12,16,20
5,10,15,20,25
```

The program is partially complete.

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Comment-First Coding
Students trained to read and interpret scaffolded comments can plan each step before coding, mirroring this task exactly.

Scaffolding to Independence
Regular practice
completing partially written programs helps learners understand structure, flow, and indentation before they tackle blank-page tasks.

Decomposition and Clarity
Breaking the task into sub-goals (calculate, format, write, close file) mirrors the decomposition process students should rehearse in class.

Q04.py - Notepad

File Edit Format View Help

```
# ---------------------------------------------------------------
# Constants
# ---------------------------------------------------------------
OUT_FILE = "Q04_OUTPUT.TXT"
COMMA = ","
LF = "\n"

# ---------------------------------------------------------------
# Global variables
# ---------------------------------------------------------------
outString = ""
maxNum = 0
num = 0
row = 0
column = 0

# ---------------------------------------------------------------
# Main program
# ---------------------------------------------------------------

# =====> Open the output file for writing


maxNum = int (input ("Enter a number: "))   # Get the number
for row in range (1, maxNum + 1):           # Going down the table
    outString = ""

    for column in range (1, maxNum + 1):    # Going across the table
        # =====> Calculate the new value

        # =====> Add a comma to all except last column

        # =====> Add a line feed to the last column

    # =====> Write the row to the file

# =====> Close the file
```

PLS Quizzes

How about a pop quiz to get students used to using the PLS? It helps reinforce key terminology, familiarises them with what's covered in the PLS and gets them used to the <notation> for arguments. E.g.

Q1: Give four examples of relational operators. (4)

Q2: Which kind of loop is used for iteration? (1)

Q3: What is the (official) name of the % operator in Python? (1)

Q4: What is the difference between a procedure and a function? (1)

Q5: What two parameters are needed to open a file for appending? (2)

Q6: What does ord(<char>) do? (1)

Assessment and feedback in both Years 10 and 11

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D1a. Y10

Q3_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:24

Q2_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:23

Q1_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:22

Q4_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:22

Q5_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:21

Q6_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:21

Q6_Y10-01-CT6_Assessment

Submit

Run

1

Assignment

Output

Q6_Y10-01-CT6_Assessment
Due Wed 26 March 2025 at 10:21

Question 6

Write a program that asks the user to enter the amount of money a bill comes to. The program should then calculate the VAT to add on (20%) and output both the amount of VAT and the new total with VAT added on. Variables should have appropriate names and output messages should be useful. (10)

D1b. Y10

Practice Question 1

Submit

Run

```python
# ------------------------------------------------------------------
# Import libraries
# ------------------------------------------------------------------
# =====> Import the required libraries
import turtle
# ------------------------------------------------------------------
# Constants
# ------------------------------------------------------------------
WIDTH = 800
HEIGHT = 600
SIDE = 90          # length of a square's side
GAP = 10           # gap between squares in the grid
# ------------------------------------------------------------------
# Global variables
# ------------------------------------------------------------------
myColours = ["red", "orange", "gold", "olive", "green", "cyan", "navy", "purple", "black"]

xPos = -145    # lower-left corner of square 1
yPos = -135

# ------------------------------------------------------------------
# Main program
# ------------------------------------------------------------------
# Window and canvas
screen = turtle.Screen()
screen.setup(WIDTH, HEIGHT)
# ------------------------------------------------------------------
# =====> Prepare the turtle (speed, pen up/down as needed)
# =====> Draw border lines A, B, C, D (black, width 3) to form a rectangle
# =====> Draw a 3x3 grid of filled squares inside the border
#        - start from (xPos, yPos) which is the lower-left corner of square 1
#        - each square has side length SIDE
#        - horizontal/vertical spacing between squares is GAP
#        - outline pensize is a random integer between 4 and 8 (inclusive)
#        - fill colours come from myColours in order, repeating if needed
#        - use begin_fill() and end_fill()
# ------------------------------------------------------------------
```

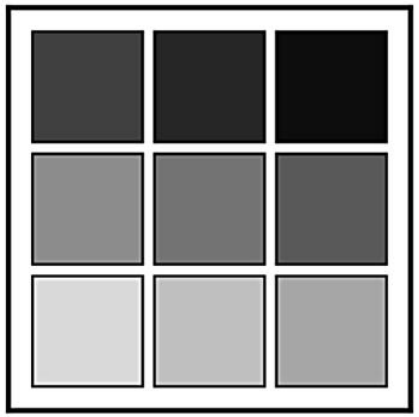
Assignment | Output

Practice Question 1
Due Fri 24 October 2025 at 12:16

Grid of Squares (Suggested time: 25 minutes)

A program is being developed to display a **3×3 grid of coloured squares** inside a black rectangular border using turtle graphics.

Here is a **greyscale image** of the output produced by a fully functional program.



D2a. Y10 – Pearson assessments 10.1–10.6

‘Real’ assessments:

- Take place in computer room using same format as actual Paper 2 exam.
- Students gain confidence and competence for the real exam.

Mark scheme:

- Colour-code mark requirements to improve understanding of how marks are gained.
- Add in student code and final code as an image for individuals to do corrections.

Question 3

A program is required to draw a pattern on the turtle graphics window.

The program must meet these requirements:

- draw a zigzag pattern
- use only the pen, not shapes
- use every colour in the data structure named ‘colourTable’, in order
- use the <turtle>.setheading(<degrees>) function to control the direction of the zigzag
- use a ‘for in range’ to reduce the need for repeated lines of code.

The output of the program is shown here.

Open the file ‘CT36_Q03_Student.py’.

Amend the code to create a program that meets the requirements above.

Do not add any additional functionality.

Save your amended code as ‘CT36_Q03_Finished.py’.

Copy and paste your code here. (10)

One mark for each up to a maximum of (10).

- Creation of a turtle with a meaningful name. (1)
- Selection of a large pen size. (1)
- Initial position chosen to allow full image to be seen. (1)
- Use of ‘for in range’ loop. (1)
- Every colour from colourTable used. (1)
- Length of leg is appropriate to show design. (1)
- Set heading used to control the zig or zag. (1)
- Heading readjusted each time through the loop. (1)
- Heading readjusted using mathematical calculation rather than ‘if’ statement. (1)
- Output matches image in question. (1)

7 marks

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D2b. Y10 diagnostic questions for formative assessment

MCQ01: For loops and the range function

What is the output from this program?

For loops and the range function MCQ01.py

```
x = 2

for i in range (1,4):
    x = x * i

print(x)
```

A. 5 B. 8 C. 12 D. 48

Starter for lesson about **for loops and in range.** Three slides. For each slide:

- each student writes answer on a Mini Whiteboard choosing from A, B, C, D
- then they fill in a solo response on google/MS form
- paired peer discussion
- fill in peer response
- teacher shows shared results to all
- class discussion to explore all answers.

Diagnosticquestions.com
Raspberry Pi Foundation
Quick Read Peer Instruction

D2a. Y10 – Pearson assessments 10.1–10.6

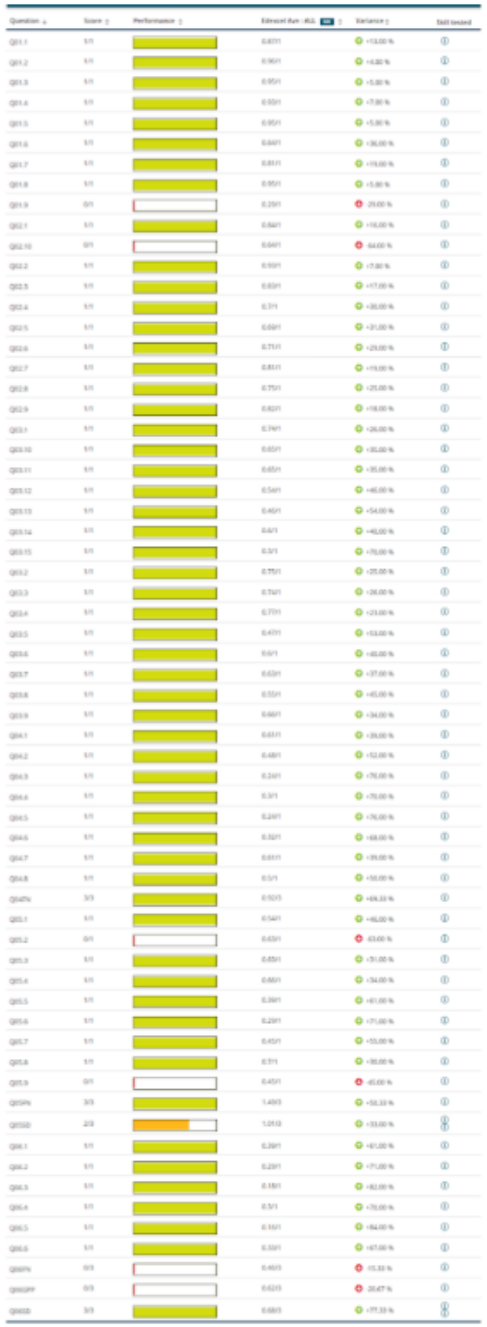
For each paper, explain what you think has happened.

What advice would you give to the Y11 student?

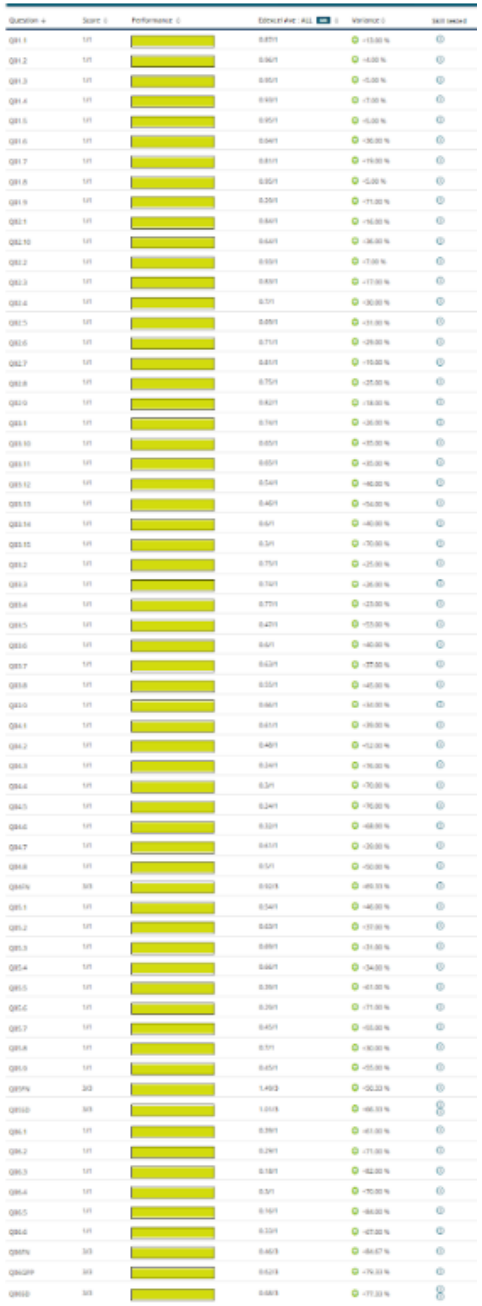
(source: ResultsPlus)

Student B - Paper 2 (38/75)

Student C - Paper 2 (65/75)



Student D - Paper 2 (75/75)



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Summary and support

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Takeaways

You will:

- be shown options for balancing the theory and programming elements of the course
- see how to organise the existing Pearson CT (Computational Thinking) units with free online platforms and resources
- be shown practical approaches for T&L of programming
- consider forms of assessment and feedback in both Years 10 and 11

Your takeaways:

1. *I will add columns for P1 and P2 to my 2025 SEF to see if there is a residual for theory or programming…*
2. *I am going to do more work on specific Parson's problem sections of code in response to the poor performance on Qu3 in 2025.*
3. ?
4. ?
5. ?

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Future webinar details

Pearson Edexcel GCSE Computer Science Professional Development training sessions

- **Preparing Students for Challenging Concepts and Principles in Paper 1**
  21 January 2026 at 16:00–17:00 GMT
- **Preparing Students for Challenging Concepts and Principles in Paper 2**
  04 March 2026 at 16:00–17:00 GMT
- **Tackling 6–Mark Questions in Paper 1**
  22 April 2026 at 16:00–17:00 BST
- **Deep Dive into Paper 2 – Questions 5 and 6**
  10 June 2026 at 16:00–17:30 BST

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Computer Science and ICT

Tim Brady

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Email: TeachingComputerScience@pearson.com

Phone: +44 (0) 344 463 2535 (Mon–Fri, 9:00–17:00)

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The following services are included as part of your qualification fees:

examWizard

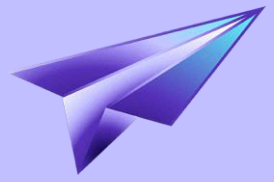
An online resource bank of past paper questions and support materials, enabling you to create customised mock exams and tests tailored to your students' needs.

ResultsPlus

An online results analysis tool that provides a detailed breakdown of students' performance in our exams, helping you identify areas for improvement and track progress effectively.

Access To Scripts

An online service that allows you to view and download your candidates' marked scripts. This tool offers transparency and insight into the marking process.



Mocks Service

We offer a paid service for mocks, including paper-based and onscreen assessments. Marked by examiners, results are uploaded to ResultsPlus for detailed analysis to aid student progression and teacher training. Note that this service is not available for all subjects; please check availability for your specific subject.

Find out more

For more professional development courses please see Pearson's Professional Development Academy.

Following this event, you will receive a survey to share your thoughts about the session. Please let us know what you'd like to see more of and what areas could be improved.

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